A Sensory Device for Obstacle Detection and Semantic Environment Recognition via SLM

Aida Araújo Ferreira
Federal Institute of Pernambuco (IFPE)
Recife, Brazil
aidaferreira@recife.ifpe.edu.br

Thiago Ewerton Barros Pereira
Federal Institute of Pernambuco (IFPE)
Recife, Brazil

Breno do nascimento santos
Federal Institute of Pernambuco (IFPE)
Recife, Brazil

Luiz Henrique Ferreira Pavão
Federal Institute of Pernambuco (IFPE)
Recife, Brazil

Alisson Rocha Ramos
Federal Institute of Pernambuco (IFPE)
Recife, Brazil

Abstract

Ensuring autonomous mobility for the visually impaired is a critical challenge, particularly in complex indoor environments. This paper presents a modular, low-cost sensory device ($150 USD) that attaches to conventional eyeglasses. Shifting the focus from simple obstacle detection to high-level spatial understanding, we introduce an approach based on Binary Occupancy Grid Mapping using a triple-array of Time-of-Flight (ToF) infrared sensors. The core innovation is the use of a local SMALL Language Model (SLM) to perform a semantic reading of the grid. By synchronizing grid updates with human odometry (step detection), the system maintains a persistent spatial memory, allowing the SLM to distinguish between architectural features such as corridors and open spaces. This camera-free, neuro-symbolic architecture ensures a Privacy-by-Design paradigm while providing high-level spatial awareness without the ethical risks of image capture.

Keywords

Visually Impaired, ToF Sensors, SLM, Ollama, Assistive Technology, Low-cost.

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1 Introduction

The autonomy and safety of people with visual impairments in urban and indoor environments remain a persistent global challenge. While recent assistive wearable technologies have improved obstacle detection through low-cost, non-visual sensing [3], safe navigation in complex architectural spaces requires more than proximity alerts. Users need to understand the surrounding spatial structure—such as distinguishing corridors from open halls—which demands higher-level semantic interpretation.

Simplified spatial representations offer a promising compromise between environmental fidelity and computational efficiency. In particular, binary occupancy grids preserve navigational semantics while remaining suitable for low-cost, resource-constrained systems [4]. This is especially relevant in developing regions: in Brazil alone, 7.9 million people experience significant visual impairment [5], underscoring the need for affordable and inclusive assistive technologies.

Existing commercial solutions fall short in this regard. High-end systems like OrCam [7] rely on continuous image capture, raising privacy concerns and costing over US$3,000 [6]. Low-cost alternatives such as WeWALK [9] enhance obstacle detection but lack explicit spatial mapping, offering limited semantic awareness.

To bridge this gap, we advance the Mark IV sensory platform [3], a US$150 device that replaces cameras with a triple-array of Time-of-Flight (ToF) sensors. In this work, we integrate a neuro-symbolic spatial reasoning layer based on a local Small Language Model (SLM). The system discretizes the environment into a 5×5 binary occupancy grid, synchronized with human odometry (step detection) to maintain temporal persistence. The SLM interprets this grid to infer architectural features—such as corridors or open spaces—and generates natural-language guidance.

By eliminating cameras and cloud processing, our architecture adheres to a Privacy-by-Design paradigm, delivering high-level spatial awareness in a low-latency, low-cost, and socially acceptable form. This work represents a novel contribution to assistive mobility, combining non-visual sensing, symbolic mapping, and local generative inference for ethical and accessible navigation.

2 Related Work and Motivation

Despite recent advances in AI-based assistive technologies, many existing solutions still present structural limitations that hinder large-scale adoption. Commercial devices such as *OrCam MyEye* provide advanced object recognition capabilities but depend on continuous image acquisition and cloud-based processing, leading to high costs, increased energy consumption, and unpredictable latency [7]. Prior analyses within the *Synesthesia Vision* project

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indicate that these dependencies significantly reduce accessibility in socioeconomically developing regions [3].

Table 1 summarizes the main characteristics of representative assistive technologies. While camera-based systems offer rich semantic perception, they introduce privacy concerns and operational constraints. Conversely, non-visual solutions, such as the white cane and WeWALK, preserve privacy but lack spatial mapping and higher-level environmental understanding.

Table 1: Comparison of assistive technologies for visually impaired users

System	Camera	AI-based	Spatial Mapping	Privacy
White Cane	No	No	No	High
WeWALK	No	Partial	No	Medium
OrCam MyEye	Yes	Yes	No	Low
Mark IV	No	Yes	Yes	High

These limitations motivate the Mark IV, which shifts from vision-based sensing to *Simplified Occupancy Grid Mapping*. Inspired by grid-based navigation approaches optimized for low-cost systems [4], the proposed architecture employs a sparse array of *Time-of-Flight* (ToF) sensors, enabling accurate spatial perception without the computational overhead and privacy risks of camera-based systems [3].

The core contribution of Mark IV is the use of a local Large Language Model (LLM) for *Semantic Grid Interpretation*, allowing the system to infer structural properties of the environment, such as corridors and open spaces. As highlighted in Table 1, this edge-based design follows a *Privacy-by-Design* paradigm, offering a scalable and socially acceptable alternative for autonomous navigation.

3 Evolution of the Synesthesia Vision Project

The Synesthesia Vision project has evolved over a decade toward balancing sensing performance, user safety, and affordability [3]. The Mark I (2015) established the initial proof of concept using an Arduino microcontroller and five ultrasonic sensors [1]. Although it successfully demonstrated non-visual spatial perception through binaural sound [1], early evaluations with volunteers revealed critical limitations, including sensory information overload due to redundant detection and latency issues caused by Bluetooth-mediated audio delivery [3].

The Mark II (2018) represented a significant refinement in usability and hardware efficiency [2]. The system transitioned to the ESP32 SoC, enabling higher processing power and integrated wireless communication in a more compact form factor [2, 3]. Based on empirical observations of user cognitive load, the sensor count was reduced from five to three to improve reaction times and directional awareness [3]. This version also introduced haptic feedback through vibration motors and simplified the mobile application to focus on supplementary contextual information, such as public transport integration in smart cities [2]. Subsequent iterations, such as the Mark III (2021), focused on miniaturization through dedicated Printed Circuit Boards (PCB), reducing bulk and improving aesthetic acceptance [3].

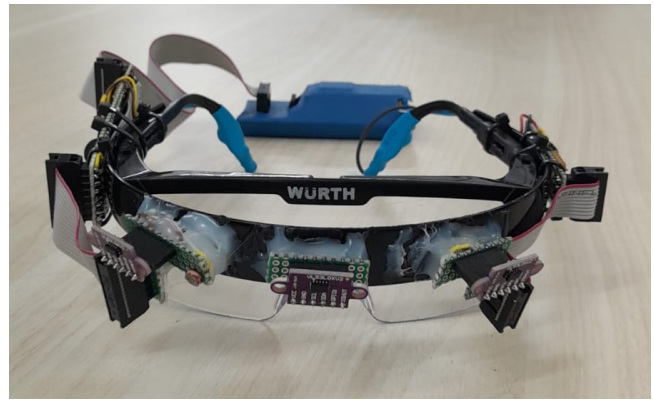

Figure 1: Mark IV prototype adopting infrared Time-of-Flight sensors and a modular design [3].

The current Mark IV (2024) represents a conceptual shift from raw sensory substitution to context-aware mobility assistance [3]. By replacing traditional ultrasonic transducers with a triple-array of infrared Time-of-Flight (ToF) sensors (VL53L0X), the system achieves millimeter-level precision, lower noise, and faster response times even in complex environments [3]. Its modular, lightweight design (approx. 95 g) integrates directly into standard eyeglass frames [3]. This allows users with residual vision to utilize the system without obstructing their remaining sight, a critical design improvement over previous versions [3]. This hardware maturity provides the technical foundation for the proposed neuro-symbolic layer and local SLM integration.

3.1 Low-Resolution Binary Occupancy Grids (BOG)

The fundamental challenge in indoor navigation for the visually impaired lies in the trade-off between environmental detail and processing latency. While high-resolution LIDAR (Light Detection and Ranging) systems generate dense point clouds capable of millimeter-level reconstruction, such data is computationally prohibitive for low-cost, battery-constrained assistive wearables. Moreover, high-resolution data is often redundant for basic trajectory planning and structural recognition. Following the "Simplified Occupancy Grid" paradigm proposed by Gonzalez-Arjona et al. [4], we implement a discretization strategy that transforms the continuous physical space into a discrete topological matrix M of 5×5 cells.

In this model, each cell $m_{i,j}$ represents a binary probability state where 1 denotes an occupied region (obstacle presence) and 0 represents a traversable free passage. This binary approach, rather than a probabilistic one (e.g., Bayesian filters), was selected to ensure maximum compatibility with the symbolic reasoning layer of the SLM. The mapping process begins at the frontal row ($i = 0$), where the conviction of occupancy is determined by a categorical thresholding function $f(d)$ applied to the Time-of-Flight (ToF) sensors. The mathematical definition of this binarization is given by:

$$m_{0,j} = \begin{cases} 1, & \text{if } \bar{d}_{ToF,j} < \tau_{proximity} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

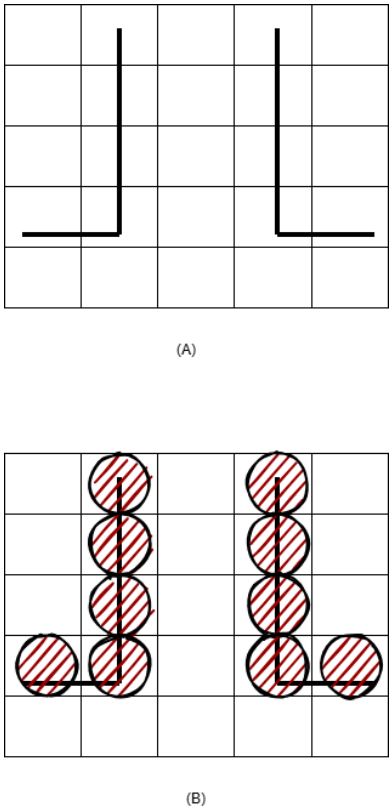

Figure 2: Spatial discretization process based on the BOBMapping paradigm: (A) Continuous representation of the structured environment; (B) Resulting binarized occupancy grid. The hatched red cells represent the "conviction of occupancy," where any detected obstacle triggers a binary state of 1, eliminating probabilistic uncertainty to optimize real-time processing for low-cost hardware [4].

where $\bar{d}_{ToF,j}$ represents the stabilized average of the last n samples for the j-th column sensor. The stabilization window is crucial to mitigate the "salt-and-pepper" noise typical of infrared sensors in environments with high ambient infrared radiation or highly reflective surfaces [3]. We defined $\tau_{proximity}$ as 75 cm for the central axis and 100 cm for the lateral axes, creating a "safety funnel" that prioritizes frontal obstructions.

The selection of a 5×5 resolution is not arbitrary; it represents a strategic balance between spatial granularity and the tokenization constraints of local Large Language Models (SLMs)1. A higher resolution (e.g., 20×20) would increase the prompt size and the model's attention overhead, potentially leading to increased inference latency. Conversely, a lower resolution would fail to distinguish between a narrow corridor and a single frontal obstacle. By utilizing 25 discrete cells, the Mark IV creates a spatial signature that can be represented as a short bitstring, optimized for the Phi-3 Mini model. This Small Language Model (SLM), featuring 3.8 billion parameters, was selected for its high-reasoning, low-footprint architecture, allowing for near-instantaneous geometric classification without cloud-based offloading.

3.2 Human Odometry and Temporal Persistence

One of the contributions of this work to the evolution of the Mark IV system is the implementation of a Step-Synchronized Temporal Memory. In standard robotic mapping approaches, occupancy grids are typically updated based on constant time intervals (dt) or high-precision wheel encoders providing odometric feedback [8]. These methods assume continuous motion estimation and precise temporal synchronization between sensors and actuators, which may not be available or reliable in wearable assistive systems.

To align the digital map with the user's proprioception, our algorithm utilizes the smartphone's Inertial Measurement Unit (IMU) to detect physical steps. This process, which we define as Human Odometry, treats each step as a discrete displacement unit in the spatial coordinate system. By leveraging the pedometer API of the mobile gateway, the system identifies a gait event and triggers a state transition in the mapping layer. Upon each detected step ($step_detected = true$), the system executes a vertical shift on the 5×5 occupancy matrix using a circular shift operation, as illustrated in the following implementation:

Algorithm 1 Step-Synchronized Occupancy Grid Exploration

```
Allocate grid matrix
Initialize all grid cells to EMPTY
Initialize track thread
Set all cell states to NOT_SENSED
InitialCell ← IdentifyInitialCell()
CurrentCell ← InitialCell
repeat
    MovDirection ← CheckNeighbors(CurrentCell)
    if MovDirection ≠ BACKWARD then
        CurrentCell ← MoveSensing(Occupancy,
    SenseData, CurrentCell, MovDirection)
        MarkTrack(Thread, CurrentCell, MovDirection)
    else
        Backtracking(Thread, CurrentCell)
    end if
until (CurrentCell = InitialCell) and (¬ExistsNeighbor(InitialCell))
```

Algorithm 2 Symbolic Neighborhood Check

```
function CheckNeighbors(Cell)
    for all dir ∈ Directions do
        if Occupancy(GetCell(dir, Cell)) = EMPTY and
    Sensed(Neighbors(GetCell(dir, Cell))) = NO then
            return dir
        end if
    end for
    return BACKWARD
end function
```

This mathematical shift, performed by the `np.roll` function, ensures that the grid functions as a "sliding window" of the environment. This mechanism addresses the "sensory amnesia" common in reactive assistive devices. For instance, a lateral wall detected by the ToF sensors three steps ago will not be forgotten; instead, it will progressively migrate from the top row ($i = 0$) to the bottom row ($i = 4$) of the matrix.

This persistence allows the system to maintain a stable semantic context even when the physical obstacle is no longer within the

sensors' immediate field of view or when the user temporarily turns their head. In practical terms, this means the local SLM (Phi-3) can still identify that the user is "inside a corridor" based on the historical data stored in the lower rows of the grid, even if the current frontal readings indicate an open path. By decoupling sensing from reporting through this temporal buffer, the Mark IV achieves a level of spatial awareness that mimics the mental mapping naturally performed by sighted individuals.

3.3 Edge Computing and Local SLM Hosting

The intelligence of the Mark IV is grounded in a local Edge Computing infrastructure, specifically engineered to uphold a Privacy-by-Design paradigm. Unlike mainstream vision-based systems that transmit high-resolution imagery to proprietary cloud servers, the Mark IV ensures that sensitive spatial telemetry remains within a controlled, local network.

We selected the Phi-3 Mini model, a Small Language Model (SLM) featuring 3.8 billion parameters. This choice was driven by the necessity of a "high-reasoning, low-footprint" architecture. While larger models like Llama-3 offer superior general knowledge, the Phi-3's optimization for logic and safety-critical tasks allows for near-instantaneous inference when hosted on our dedicated laboratory server, equipped with an NVIDIA RTX series GPU and the Ollama inference framework.

The communication pipeline is structured through a low-latency REST protocol over an intranet link, executing a deterministic four-stage flow:

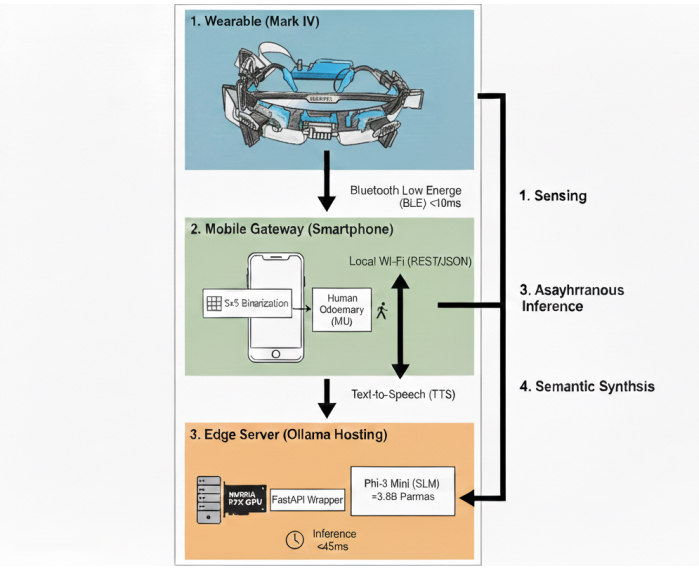

Figure 3: Mark IV Neuro-Symbolic Pipeline: Data flow starts from ToF sensing (1), passes through step-synchronized binarization in the mobile gateway (2), followed by local SLM inference (3), and concluding with semantic synthesis for auditory feedback (4). The architecture ensures privacy and low latency by keeping all data within the local network.

4 Results and Conclusion

The evaluation of the Mark IV architecture was conducted through a hybrid validation process. Although the wearable hardware and its physical ergonomics had been previously validated through testing with visually impaired users, the specific performance of neurosymbolic mapping and local SLM integration was evaluated under controlled laboratory conditions. This approach allowed for precise baseline measurement of Round-Trip Time (RTT) and grid persistence, without the safety risks associated with early-stage algorithmic testing.

Table 2: System Validation: Semantic Environment Recognition and Grid Persistence

Real Scenario	ToF Readings (L, C, R)	Context (Code)	Phi-3 Instruction
1. Start (Clear)	300, 300, 300	`WIDE_AREA`	"Wide area ahead. Proceed."
2. Frontal Block	300, 50, 300	`FRONTAL_BLOCK`	"Obstacle ahead. Please stop."
3. Left Side Wall	45, 300, 300	`LEFT_WALL`	"Wall detected on your left."
4. Inside Corridor	45, 300, 45	`CORRIDOR`	"You are inside a corridor."
5. Exit (Persistence)	300, 300, 300	`CORRIDOR`	"Exiting the corridor now."

Quantitative results recorded an average RTT of 185 ms. Within this controlled environment, the system successfully simulated five navigation archetypes. As shown in Table 2, the occupancy grid effectively maintained the "Corridor" context for three steps after sensor clearing—a direct result of human odometry. These results suggest that the system can operate effectively at a standard human walking pace (1.2 m/s), providing a safety margin of approximately 1.5 meters.

In conclusion, Mark IV demonstrates that a neurosymbolic architecture without expensive sensors can provide semantic awareness. By combining binary occupancy grids with local SLM inference, we have established a scalable paradigm for assisted mobility. Future work will prioritize extensive field testing with visually impaired participants to evaluate the system's robustness against real-world environmental noise and dynamic obstacles. Additionally, we aim to implement sensor fusion between ToF and ultrasonic data to mitigate specular reflections, while exploring the integration of this semantic layer with IoT-enabled smart city infrastructures to provide context-aware information on public transport arrivals.

References

[1] Michael Barney, Jonathan Kilner, Gilmar Brito, Aida Araújo, and Meuse Nogueira. 2017. Sensory glasses for the visually impaired. In *Proceedings of the 14th International Web for All Conference (W4A '17)*. ACM, Perth, Australia. doi:10.1145/3058555.3058584

[2] Gilmar G. de Brito, Aida Ferreira Araújo, Thiago Santos, and Ronaldo R. B. Aquino. 2019. Synesthesia Vision: Low-cost Smart Glasses for Blind Navigation. In *Proceedings of the 16th International Web for All Conference (W4A '19)*. ACM, San Francisco, CA, USA. doi:10.1145/3315002.3332446

[3] Aida Araújo Ferreira, Carlos Alberto Leal de Barros, Thiago Ewerton Barros Pereira, Maria Francielly A. de Santana, Isaque D. S. Silva, Felipe S. de Oliveira, Gilmar G. de Brito, Ioná M. B. R. Barbosa, Vânia Soares de Carvalho, Dalton Cézane Gomes Valadares, Ryan Vinicius Santos Morais, and Ronaldo R. B. de Aquino. 2026. Modular Smart Glasses for Real-Time Obstacle Detection in Blind

Navigation. *IEEE Internet of Things Magazine* (2026), 1–8. doi:10.1109/MIOT.2025.3643746

[4] David Gonzalez-Arjona, Alberto Sanchez, Fernando López-Colino, Angel de Castro, and Javier Garrido. 2013. Simplified Occupancy Grid Indoor Mapping Optimized for Low-Cost Robots. *ISPRS International Journal of Geo-Information* 2, 4 (2013), 959–977. doi:10.3390/ijgi2040959

[5] Instituto Brasileiro de Geografia e Estatística. 2023. Censo Demográfico 2022: Pessoas com deficiência e limitações funcionais. https://www.ibge.gov.br. Accessed: Dec. 23, 2026.

[6] Paul McCullagh, Victoria Bellotti, Thomas Erickson, and Wendy A. Kellogg. 2020. Privacy Considerations of the Visually Impaired with Camera-Based Assistive Technologies. *ACM Transactions on Accessible Computing* (2020).

[7] OrCam Technologies. 2025. OrCam MyEye. https://www.orcam.com/en/myeye/. Accessed: Jan. 2026.

[8] Sebastian Thrun, Wolfram Burgard, and Dieter Fox. 2005. *Probabilistic Robotics*. MIT Press.

[9] WeWALK. 2025. WeWALK Smart Cane. https://wewalk.io/en/. Accessed: Jan. 2026.